

Application of Gaussian Processes

① Discriminate Analysis (Classification)

N 个 Categories, 分别为 $C_1, C_2, \dots, C_N$. Data 为 $x_1, \dots, x_n$, 且 $x_k \sim C_i \sim N(\mu_i, \Sigma_i)$

已知

进行分类时, 一个自然的想法为: Distance: $\min_i (x_k - \mu_i)^T (x_k - \mu_i)$

建立在 Σ_i 差不多大的基础上

能否换为: $\min_i (x_k - \mu_i)^T \Sigma_i^{-1} (x_k - \mu_i)$ $\rightarrow$ Mahalanobis Distance

马氏距离

没有考虑先验知识

加上 Bayesian: $\max_i P(C_i | x_k) = \max_i \frac{P(x_k | C_i) P(C_i)}{P(x_k)}$

$$= \max_i \frac{P(x_k | C_i) P(C_i)}{\sum_j P(x_k | C_j) P(C_j)}$$

$$\Leftrightarrow \max_i \left(\log P(C_i) + \log P(x_k | C_i) - \log \left(\sum_j P(x_k | C_j) P(C_j) \right) \right)$$

$$\text{即 } \log P(x_k | C_i) = -\frac{1}{2} (x_k - \mu_i)^T \Sigma_i^{-1} (x_k - \mu_i)$$

Bayesian Factor

进一步, 若假定 $P(C_i) = \frac{1}{N}$, 可写为:

$$\max_i \frac{\exp \left(-\frac{1}{2} (x_k - \mu_i)^T \Sigma_i^{-1} (x_k - \mu_i) \right)}{\sum_j \exp \left(-\frac{1}{2} (x_k - \mu_j)^T \Sigma_j^{-1} (x_k - \mu_j) \right)}$$

Nearest Center Classification

更进一步, 假定 $\Sigma_i = \Sigma$.

$$\max_i \frac{\exp \left(-\frac{1}{2} (x_k - \mu_i)^T \Sigma^{-1} (x_k - \mu_i) \right)}{\sum_j \exp \left(-\frac{1}{2} (x_k - \mu_j)^T \Sigma^{-1} (x_k - \mu_j) \right)}$$

$$\Rightarrow \max_i \frac{\exp(x_k^T \beta_i + \gamma_i)}{\sum_j \exp(x_k^T \beta_j + \gamma_j)}$$

Softmax

② Gaussian Process Regression

$(x_1, \dots, x_n)$ 预测 $\rightarrow x_{n+1}$. 常见的回归为线性回归: $x_{n+1} = \sum_k \alpha_k x_k$

1. Regard $(x_1, \dots, x_n, x_{n+1})$ As Gaussian Process

2. Estimation of Hyperparameters of this GP (mean, covariance)

$$E(x_{n+1} | x_1, \dots, x_n) = \mu_{n+1} + C_{(1:n), n+1} C_{(1:n), (1:n)}^{-1} ((x_1, \dots, x_n) \cdot (\mu_1, \dots, \mu_n))$$

$$C_{(1:n), (1:n)}(i, j) = E(x_i x_j) \rightarrow k(x_i, x_j) = \exp \left(-\frac{\|x_i - x_j\|^2}{\sigma_{ij}^2} \right)$$

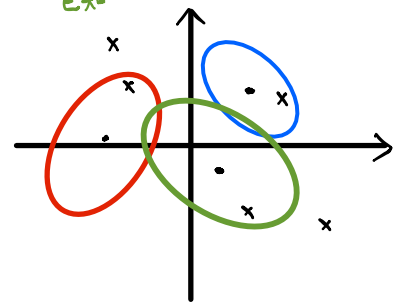
Radial Basis

Finance vs GP (Brown Motion)

$B(t)$: GP. Independent Increment. $B(t) - B(s) \sim N(0, \sigma^2(t-s))$

用 $B(t)$ 解释 Stock Price. 但是 Price 不可能有负的 $\Rightarrow B(t) \rightarrow \exp(\mu t + B(t)) \geq 0$

Geometric Brown Motion



Ito (1944) : $f(t, B(t)) \Rightarrow df(t, B(t)) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial B} dB(t)$ ✗

$dB(t) \simeq B(t+dt) - B(t) \sim N(0, \sigma^2 dt)$

又 $E(dB(t)) = 0$, $E(dB(t))^2 = \sigma^2 dt \Rightarrow dB(t) \sim \sigma \sqrt{dt}$

实际上, $df(t, B(t)) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial B} \underbrace{dB(t)}_{\sqrt{dt}} + \frac{1}{2} \frac{\partial^2 f}{\partial B^2} \underbrace{dt}_{dt}$ (Ito Formula)

e.g. $\int_0^1 B(t) dB(t) = \int_0^1 d(\frac{1}{2} B^2(t)) = \int_0^1 (B(t) dB(t) + \frac{1}{2} dt)$
 $= \frac{1}{2} B^2(t) \Big|_0^1 + \frac{1}{2} t \Big|_0^1$
 $= \frac{1}{2} B^2(1) - \frac{1}{2}$

e.g. $S(t) = \exp(\mu t + \sigma^2 B(t))$

$dS(t) = d \exp(\mu t + \sigma^2 B(t)) = (d \exp(\mu t)) \exp(\sigma^2 B(t)) + \exp(\mu t) d(\exp \sigma^2 B(t))$
 $= \exp(\mu t) \cdot (\mu dt) \exp(\sigma^2 B(t)) + \exp(\mu t) \left[\exp(\sigma^2 B(t)) \sigma^2 dB(t) + \frac{\sigma^4}{2} \exp(\sigma^2 B(t)) dt \right]$
 $= \exp(\mu t + \sigma^2 B(t)) \left[\left(\mu + \frac{\sigma^4}{2} \right) dt + \sigma^2 dB(t) \right]$
 $= S(t) \left[\left(\mu + \frac{\sigma^4}{2} \right) dt + \sigma^2 dB(t) \right]$

(Black - Scholes - Merton)

Option Pricing

期权 $\longrightarrow$ Hedge 对冲

$V(t, S(t)) - \alpha S(t) \triangleq P(t)$



$dP(t) = r P(t) dt$
 Interest Rate 银行利率

$dP(t) = dV(t, S(t)) - \alpha dS(t)$
 $= \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS(t) + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} (\sigma^4 S^2(t) dt) - \alpha dS(t)$

$\alpha = \frac{\partial V}{\partial S}$ (Delta Hedge)

$\left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^4 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt = r \left(V - \frac{\partial V}{\partial S} S \right) dt$

Boundary Condition

即 $\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^4 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - r \cdot V = 0$ Black - Scholes Equation $V(T) = \text{ReLU}(S(T) - K)$